GORENSTEINNESS OF A SCHEME IS LOCAL FOR THE SYNTOMIC TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0C01 (sources/stacks-duality/pinned-original-source.tex line 6436).

Allowed standard prerequisites (editorial).

- Local Noetherianity is local for the fppf topology. Descent, Lemma labelled descent-lemma-Noetherian-local-fppf, cited at proof lines 6456-6460. Allows assuming the schemes locally Noetherian.
- Gorensteinness is preserved by local complete intersection morphisms. Same chapter, Lemma labelled lemma-gorenstein-lci, cited at proof lines 6460-6462. Gives the forward direction.
- Descent of Gorensteinness along a flat local homomorphism. Dualizing Complexes, Lemma labelled dualizing-lemma-flat-under-gorenstein, cited at proof lines 6466-6468. Gives the converse direction.

Source reference. Original lemma 034C: sources/stacks-chow/tags.txt, source lines 4087–4087. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6444 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BVA: sources/stacks-duality/pinned-original-source.tex, source lines 6424–6424. Same-chapter pointer into the author environment lemma at source line 6424; invoked at proof line 6447 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0BJL: sources/stacks-chow/tags.txt, source lines 14529–14529. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 6453 of the counted proof. It is

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Lemma 0.1. *The property $\mathcal{P}(S) =$"S is Gorenstein" is local in the syntomic topology.*

Proof. Let $\{S_i \to S\}$ be a syntomic covering. The scheme S is locally Noetherian if and only if each S_i is Noetherian, see Descent, Lemma 034C. Thus we may now assume S and S_i are locally Noetherian. If S is Gorenstein, then each S_i is Gorenstein by Lemma 0BVA. Conversely, if each S_i is Gorenstein, then for each point $s \in S$ we can pick i and $t \in S_i$ mapping to s. Then $\mathcal{O}_{S,s} \to \mathcal{O}_{S_i,t}$ is a flat local ring homomorphism with $\mathcal{O}_{S_i,t}$ Gorenstein. Hence $\mathcal{O}_{S,s}$ is Gorenstein by Dualizing Complexes, Lemma 0BJL. □